

SPG-Net: A Structure-aware Spatiotemporal Graph Neural Network for Electrocardiogram-based Myocardial Infarction Localization

Abstract—Myocardial infarction remains a leading cause of cardiovascular mortality, making its early detection and localization via electrocardiograms (ECGs) critical for timely intervention. Despite advances in deep learning-based ECG analysis, two key challenges persist: (1) Severe class imbalance biases models toward dominant categories, hindering the learning of discriminative patterns for underrepresented subtypes (e.g., posterior, extensive anterior infarctions). (2) The inability to adequately model complex spatiotemporal dependencies in ECGs limits the effectiveness of current approaches. To address these challenges, we propose SPG-Net, a novel structure-aware spatiotemporal graph network that integrates clinical priors with dynamic spatiotemporal dependencies of ECG for MI localization. The framework extracts high-resolution lead-wise features and constructs anatomically grounded graphs using class-specific lead contribution maps. A dedicated graph attention module integrates static anatomical priors, inter-lead spatial interactions, and temporal dependencies to generate ECG feature representations that preserve physiological consistency while accommodating individual variability. Furthermore, we design a decoupled classification mechanism to reduce interference from overlapping labels, significantly improving minority-class recognition. Comprehensive experiments on PTB and PTB-XL datasets demonstrate SPG-Net’s superior overall performance and significantly enhanced minority-class recognition, validating its efficacy in class-imbalanced MI localization.

Index Terms—Myocardial infarction, electrocardiogram analysis, class imbalance, graph neural networks, spatiotemporal modeling.

I. INTRODUCTION

Myocardial infarction (MI), primarily induced by acute coronary artery occlusion, remains one of the leading causes of cardiovascular mortality worldwide. According to recent projections, cardiovascular diseases—including MI—are expected to cause over 20.5 million deaths in 2025, rising to 35.6 million by 2050, driven largely by global population aging and the persistent burden of atherosclerotic disease [1]. Timely detection and anatomical localization of MI are critical for clinical intervention and patient prognosis.

Electrocardiography (ECG) serves as a first-line diagnostic tool owing to its non-invasiveness, cost-effectiveness, and real-time interpretation capability. The standard 12-lead ECG captures comprehensive electrophysiological activity and enables regional localization of infarcted areas. However, the interpretation of ECG signals remains challenging due to complex spatial correlations among leads, subtle waveform variations, and heterogeneous pathological manifestations. Overlapping features across MI subtypes further hinder manual analysis and

consistency. Recent advances in deep learning have enabled promising progress in automated ECG analysis, particularly for arrhythmia detection and MI classification [2]. Models based on convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and attention mechanisms have demonstrated strong capabilities in feature extraction and classification under limited supervision [3]. Emerging architectures such as Transformers and graph neural networks (GNNs) have further improved model expressiveness and generalization across patient populations [4], [5]. Despite these developments, two major challenges remain unresolved: (1) Severe class imbalance biases models toward dominant categories, hindering the learning of discriminative patterns for underrepresented subtypes [6]; and (2) The inability to adequately model complex spatiotemporal dependencies in ECGs limits the effectiveness of current approaches [7], [8].

To overcome these issues, a novel framework—Structure-aware Prior-guided Graph Network (SPG-Net)—is proposed for MI localization. It integrates clinically guided lead-level priors with spatiotemporal graph modeling to improve class-specific discrimination and structural interpretability. The contributions are four-fold:

- Class-aware structural graphs are constructed from lead contribution maps (LCMs) derived from model responses, providing meaningful anatomical priors to guide inter-lead spatial relationship modeling.
- A Graph-Spatiotemporal Cross Attention (GSCA) module is developed to jointly model anatomical priors, dynamic lead interactions, and long-range temporal dependencies for effective ECG representation learning.
- A Class-Specific Adaptive Classifier (CSAC) is employed to decouple multi-label classification paths, enhancing the recognition of minority MI classes and reducing inter-class interference.
- Comprehensive evaluations on the PTB and PTB-XL datasets demonstrate that SPG-Net consistently outperforms state-of-the-art methods in both global and class-specific metrics, validating its effectiveness for ECG-based MI localization under imbalanced data conditions.

II. RELATED WORK

A. Spatiotemporal Modeling in ECG

Effective extraction of spatiotemporal features from ECG signals is crucial for accurately localizing MI, as it enables

models to learn complex inter-lead relationships and temporal evolution of pathological patterns. Traditional deep learning models often struggle to capture these dependencies, especially when faced with heterogeneous infarction regions and patient-specific variations.

To address these limitations, various architectures have been proposed. Han et al. integrated residual and LSTM modules in their model and used Grad-CAM visualizations to capture spatial dependencies across ECG leads [9]. Liu et al. introduced SRTNet, which performs ECG analysis in three stages—Scan, Read, and Think—mimicking human reasoning to enhance localization accuracy [10]. Spa-Tem MI combined spatial and temporal attention using matrix-based representations, allowing the model to differentiate frequency-band evolution across leads [11]. Guo et al. proposed a Lead-Grouped Multi-Stage architecture that embeds lead position and performs stage-wise reordering to enhance lead interaction modeling [12].

Despite their contributions, most of these methods rely on fixed structural assumptions or limited fusion mechanisms. They often construct static lead representations, which restrict adaptability to diverse infarction patterns. Moreover, the lack of dynamic modeling hinders their ability to learn long-range temporal dependencies and individualized anatomical variations, which are essential for robust and physiologically accurate MI localization.

B. Classification and Localization of MI

Class imbalance poses a significant challenge in ECG-based MI classification, where minority MI subtypes are often underrepresented, leading to degraded performance and bias in clinical predictions. To mitigate this, various strategies at the data, loss, and ensemble levels have been explored.

Sampling-based approaches like SMOTE-Tomek have been combined with CNN-LSTM architectures to rebalance data distributions and enhance minority class recognition [13], [14]. Loss function reweighting strategies have also been introduced. Zubair et al. used cost-sensitive learning to guide CNNs in focusing on rare anomalies [15], while Cai et al. developed a dynamic batch weighting loss to amplify the representation of minority classes [16]. Dey et al. proposed hierarchical feature pruning to retain salient patterns while minimizing redundancy, improving generalizability across classes [17].

Beyond loss and data-level methods, ensemble and transfer learning techniques have been explored to improve robustness. Abbas et al. presented a deep ensemble framework with sample rebalancing and normalization, achieving better multi-label MI classification [18]. Akter et al. integrated electronic health records (EHR) with class-aware sampling and fair feature selection to reduce disparity across subtypes [19].

While these approaches improve performance under imbalance, they primarily manipulate input distributions or loss gradients without fundamentally enhancing the representational capacity of the model. As a result, class-specific features may still be poorly learned, limiting the precision of predictions, especially in clinically critical minority classes.

III. PROPOSED METHOD

The overall architecture of SPG-Net is shown in Fig. 1. SPG-Net is a structure-aware spatiotemporal graph neural network for MI localization using 12-lead ECG signals. It first encodes ECG data via a Hierarchical Residual Feature Pyramid (HRFP), followed by a Dynamic Biased Weights (DBW) module that improves minority-class sensitivity and generates lead contribution maps (LCMs). These maps are aggregated across training samples to build class-aware structural graphs, which serve as fixed anatomical priors throughout training. During each iteration, ECG data are re-encoded via HRFP and processed by a Graph-Spatiotemporal Cross Attention (GSCA) module, which integrates these priors with dynamic spatial and temporal modeling. A Class-Specific Adaptive Classifier (CSAC) finalizes predictions with improved minority-class discrimination.

A. Lead-Aware Structural Prior Construction

To embed class-sensitive anatomical priors into the network, SPG-Net constructs structural graphs that represent collaborative lead activation patterns for different MI subtypes. This process begins with temporal feature encoding via HRFP, followed by supervised classification with DBW, and culminates in class-aware structural graph construction using lead contribution analysis.

1) *Hierarchical Residual Feature Pyramid (HRFP)*: To extract multi-scale temporal features from 12-lead ECGs and compress dimensionality, SPG-Net employs a Hierarchical Residual Feature Pyramid (HRFP). As depicted in Fig. 2, the architecture comprises a transposed convolutional layer, four sequentially stacked residual blocks, and an adaptive max pooling layer. Each residual block incorporates multi-scale 1D convolutions and Squeeze-and-Excitation (SE) attention mechanisms to capture temporal dynamics across different scales. Let the input be $\mathbf{X} \in \mathbb{R}^{B \times 12 \times T}$. The forward process of HRFP can be expressed as:

$$\mathbf{H}_{\text{HRFP}} = \text{AMP}(\text{RB}_4(\text{RB}_3(\text{RB}_2(\text{RB}_1(\text{TConv}(\mathbf{X})))))) \quad (1)$$

where TConv, RB_i , and AMP denote the transposed convolution, the i -th residual block, and the adaptive max pooling operation, respectively.

These residual blocks downsample the temporal dimension by half and expands the channel depth progressively ($4 \rightarrow 8 \rightarrow 16 \rightarrow 32$). A final adaptive max pooling layer compresses the temporal axis to size 1, producing a lead-wise representation of shape $\mathbb{R}^{B \times 32 \times 1}$ for downstream modules.

All residual blocks include residual connections with 1×1 convolutions for dimension alignment, ensuring stable feature fusion. Compared to conventional shallow CNNs, HRFP enhances spectral sensitivity and waveform detail preservation, providing robust lead-level features—especially beneficial for detecting minority-class infarctions.

2) *Dynamic Biased Weights (DBW)*: To address the class imbalance problem across MI subtypes, SPG-Net adopts a Dynamic Biased Weights (DBW) mechanism. The model is formulated as seven parallel binary classification tasks, each

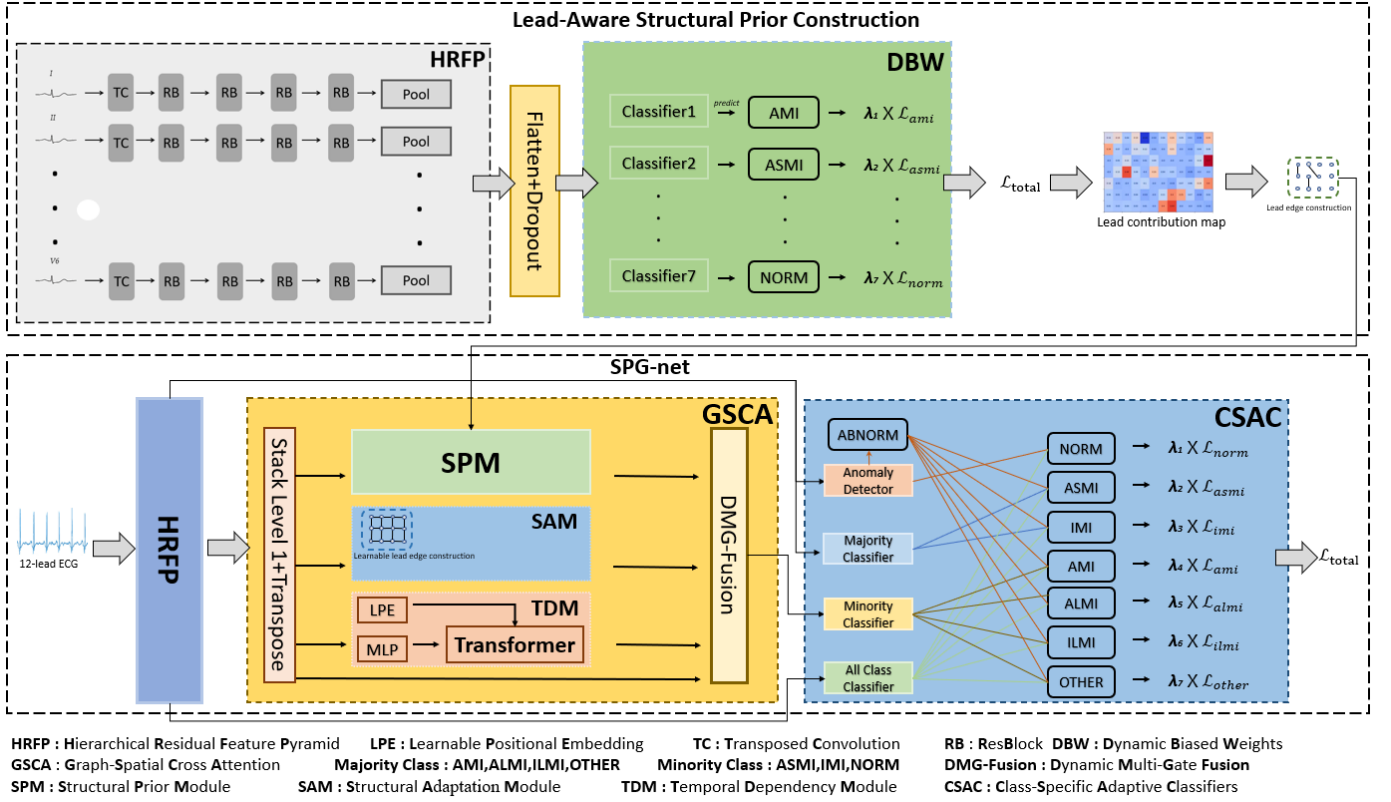


Fig. 1: The overall architecture of SPG-Net is illustrated.

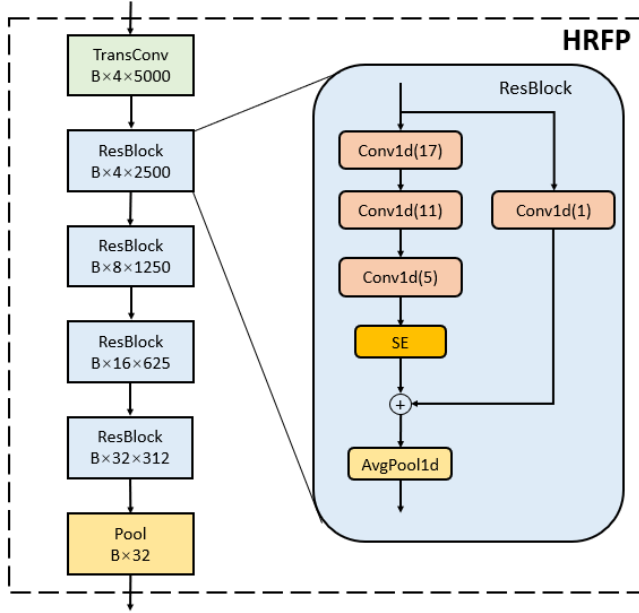


Fig. 2: Overview of HRFP.

associated with a specific MI category (e.g., anterior, inferior, lateral, etc.). A fixed class-specific weight λ_i is applied to each

binary cross-entropy loss:

$$L_{\text{total}} = \sum_{i=1}^7 \lambda_i \cdot \mathcal{L}_{\text{BCE}}^{(i)} \quad (2)$$

where $\mathcal{L}_{\text{BCE}}^{(i)}$ denotes the loss for the i -th classification branch, and λ_i adjusts the gradient contribution of each class. λ_i is selected by grid search within [1.0, 3.0] to optimize class-wise F1. This weighted optimization enhances feature learning for underrepresented classes and improves the quality of LCMs used in graph construction.

3) *Class-Aware Structural Graph Construction*: To incorporate class-sensitive anatomical priors, SPG-Net constructs class-aware structural graphs from lead contribution maps (LCMs) after DBW-guided classification. LCMs are derived by computing the gradient of the prediction score with respect to each lead's features, yielding a saliency-based contribution map that reflects the importance of each lead in the final decision. Each LCM consists of a 12-dimensional contribution vector, where each element corresponds to the saliency value of one ECG lead. In essence, the Lead Contribution Map quantifies the gradient-based importance of each ECG lead for class-specific predictions.

As depicted in Fig. 3, based on the obtained lead contribution maps (LCMs), the graph construction process involves: (1) extracting lead contributions for minority classes, (2) selecting salient leads, and (3) defining inter-lead edges using Pearson

correlation, which models linear co-activation between ECG leads in a physiologically consistent manner.

Given a trained HRFP encoder, we compute the class-specific lead response vector $C^{(k)} \in \mathbb{R}^{12}$, where $C_i^{(k)}$ denotes the average contribution of lead i to class k . The ECG leads are represented as a node set $V = \{v_1, v_2, \dots, v_{12}\}$. For each class k , a prior graph is defined as $G_{\text{prior}}^{(k)} = (V_{\text{select}}^{(k)}, E^{(k)})$, where $V_{\text{select}}^{(k)} \subseteq V$ includes the most responsive leads selected by thresholding on $C^{(k)}$, and $E^{(k)}$ captures pairwise co-activation. Let u_i be the contribution vector of lead i , and $u_i^{(k)}$ its value in sample k . The Pearson correlation coefficient between leads i and j is computed as:

$$r_{ij} = \frac{\sum_{k=1}^N (u_i^{(k)} - \bar{u}_i)(u_j^{(k)} - \bar{u}_j)}{\sqrt{\sum_{k=1}^N (u_i^{(k)} - \bar{u}_i)^2} \cdot \sqrt{\sum_{k=1}^N (u_j^{(k)} - \bar{u}_j)^2}} \quad (3)$$

If $r_{ij}^{(k)} > \theta$, an edge is added: $E^{(k)} = \{(i, j) \mid r_{ij}^{(k)} > \theta\}$. The threshold θ is empirically set to 0.95 to preserve only strong, physiologically meaningful correlations among ECG leads. Thus, $G_{\text{prior}}^{(k)}$ captures co-activated lead relationships specific to class k .

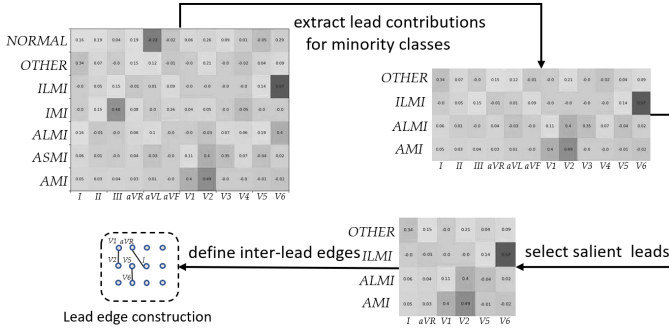


Fig. 3: Class-aware structural graph construction process.

The full set $\{G_{\text{prior}}^{(k)}\}$ forms the structural prior used by the GSCA module. These model-derived graphs offer a label-free, physiologically consistent prior that enhances minority-class representation and supports class-specific spatial modeling in GSCA.

B. Graph-Spatiotemporal Cross Attention Module (GSCA)

Given any ECG signal $\mathbf{X}$, it is first processed by the HRFP module to obtain the corresponding lead-wise feature representation $\mathbf{H}_{\text{HRFP}}$. Building upon this representation, we design a Graph-Spatiotemporal Cross Attention (GSCA) module to address three key challenges in myocardial infarction localization: (i) spatial inconsistency among ECG leads, (ii) individual variability in structural activation patterns, and (iii) limited capacity for modeling long-range temporal dependencies.

GSCA consists of three complementary submodules: the Structural Prior Module (SPM), the Structural Adaptation Module (SAM), and the Temporal Dependency Module (TDM). These branches extract spatial and temporal information from different perspectives, and their outputs are fused to form a unified spatiotemporal representation for classification.

1) *Structural Prior Module (SPM)*: The Structural Prior Module enhances the model's ability to represent MI regions that exhibit clear anatomical localization, especially for minority classes. Conventional convolutional or fully connected architectures lack the capacity to encode spatial topology among ECG leads, particularly when data is sparse. To address this limitation, we adopt graph convolutional networks (GCNs) to propagate lead features over a class-aware prior graph. The input to the GCN is the Class-Aware Structural Graph introduced in the previous subsection, which captures domain-specific co-activation relationships among ECG leads.

For each MI category, a structural prior graph $G_{\text{prior}}^{(k)} = (V_{\text{select}}^{(k)}, E^{(k)})$ is used. Here, $V_{\text{select}}^{(k)}$ denotes selected critical leads and $E^{(k)}$ encodes edges based on statistical lead correlation patterns. Given the lead-level feature input $H^{(l)}$, graph convolution is applied as follows:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}) \quad (4)$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-loops, $\tilde{D}$ is the degree matrix, $W^{(l)}$ is the trainable weight matrix of layer l , and $\sigma(\cdot)$ is the activation function, $H^{(l)} \in \mathbb{R}^{|V| \times d_l}$ denotes the node feature matrix at layer l , where each row corresponds to a selected lead and its feature embedding. The initial input $H^{(0)}$ is obtained from the HRFP module and represents lead-wise features extracted for each sample.

This spatial modeling strategy allows the network to preserve clinical structural knowledge during training, reinforcing collaborative patterns among informative leads. As a result, SPM improves MI localization performance in anatomically distinct and data-scarce categories.

2) *Structural Adaptation Module (SAM)*: While structural dependencies between ECG leads can be modeled through prior graphs, in real-world ECG acquisition scenarios, significant inter-individual variations exist in lead responses. Moreover, certain infarction regions (e.g., the posterior wall) lack clear anatomical projections in the standard lead layout. This structural heterogeneity makes it difficult for static graphs to effectively capture lead-wise co-activation patterns across diverse patient profiles. To enhance the model's capacity for individualized structural modeling, we propose the Structural Adaptation Module (SAM), which dynamically constructs a learnable sample-adaptive graph topology. Built upon a graph neural network framework, SAM supports end-to-end differentiable feature propagation across dynamically determined lead connections.

We implement this module using the GraphSAGE architecture. Each of the 12 ECG leads is treated as a graph node v_i , forming the node set: $V = \{v_1, v_2, \dots, v_N\}$, where $N = 12$. A learnable adjacency matrix $A_{\text{learnable}} \in \mathbb{R}^{N \times N}$ is introduced to model pairwise lead interaction strengths. A row-wise softmax normalization is applied to generate the soft connectivity weights:

$$A_{\text{soft}} = \text{Softmax}(A_{\text{learnable}}, \text{dim} = -1), \quad (5)$$

where each element $a_{ij} \in (0, 1)$ of A_{soft} represents the dynamic interaction weight between leads i and j . Non-zero entries in A_{soft} define the edge set $E \subseteq V \times V$, thereby forming a fully differentiable, sample-specific graph $G = (V, E)$.

During feature propagation, GraphSAGE performs graph sampling and aggregation to update node embeddings. Let $h_i^{(l)}$ denote the embedding of lead i at layer l . The update rule is given by:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \left[h_i^{(l)} \parallel \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} a_{ij} \cdot h_j^{(l)} \right] \right), \quad (6)$$

where $\mathcal{N}(i)$ denotes the neighborhood of node i , $W^{(l)}$ is a learnable weight matrix, $\sigma(\cdot)$ is a non-linear activation function, $\parallel$ denotes feature concatenation, a_{ij} is the edge weight from A_{soft} . This module is independent of any static structural prior and dynamically constructs lead-level message-passing paths for each input sample.

3) *Temporal Dependency Module (TDM)*: The Temporal Dependency Module models long-range dynamics in ECG sequences using a Transformer encoder with relative positional encoding. The input is a tensor $\mathbf{X} \in \mathbb{R}^{12 \times T \times d}$, where T is the temporal length and d is the feature dimension. For each lead l , the sequence $\mathbf{X}_l \in \mathbb{R}^{T \times d}$ is processed independently.

A learnable relative position matrix $\mathbf{R} \in \mathbb{R}^{(2T-1) \times d}$ is used to enhance temporal representations:

$$\tilde{\mathbf{X}}_i = \mathbf{X}_i + \sum_{j=1}^T \mathbf{R}[i - j + T - 1] \quad (7)$$

where $\mathbf{R}[i - j + T - 1]$ encodes the offset between time-steps i and j . Self-attention is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \quad (8)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ are query, key, and value projections and d_k is the key dimension. This module improves temporal sensitivity and enables each time point to attend globally across the lead sequence.

4) *Fusion Mechanism*: To integrate structural representations from GSCA with semantic features from HRFP, we introduce a residual identity path that passes the original lead-level features $\mathbf{H}_{\text{HRFP}} \in \mathbb{R}^{12 \times d}$ directly into the fusion stage.

The three branches of GSCA output refined features

$$\alpha_i = \text{MLP}_i(\mathbf{H}_{\text{HRFP}}), \quad i \in \{\text{SPM}, \text{SAM}, \text{TDM}\} \quad (9)$$

Each MLP contains a linear layer with ReLU and Sigmoid activations, ensuring outputs lie in $(0, 1)$. The fused representation is computed as:

$$\mathbf{H}_{\text{fused}} = \alpha_{\text{SPM}} \odot \mathbf{H}_{\text{SPM}} + \alpha_{\text{SAM}} \odot \mathbf{H}_{\text{SAM}} + \alpha_{\text{TDM}} \odot \mathbf{H}_{\text{TDM}} + \mathbf{H}_{\text{HRFP}} \quad (10)$$

where $\odot$ denotes element-wise multiplication, $\mathbf{H}_{\text{SPM}}$ refined features from the *Structural Prior Module*, $\mathbf{H}_{\text{SAM}}$ refined features from the *Structural Adaptation Module* and $\mathbf{H}_{\text{TDM}}$ refined features from the *Temporal Dependency Module*. This

fusion method dynamically balances contributions from each module and preserves interpretability. The final $\mathbf{H}_{\text{fused}}$ is forwarded to CSAC for classification.

C. Class-Specific Adaptive Classifiers (CSAC)

To address label imbalance and semantic overlap in MI localization, SPG-Net introduces a Class-Specific Adaptive Classifier (CSAC) to decouple the seven-class task into multiple expert heads. The classification heads consist of: (i) a binary classifier $\mathbf{z}^{\text{HC}} \in \mathbb{R}^2$ that separates normal (Healthy Control, HC) from abnormal samples; (ii) a binary classifier $\mathbf{z}^{\text{M}} \in \mathbb{R}^2$ for majority-class MI types, namely Inferior wall MI (IMI) and Posterior wall MI (ASMI); and (iii) a four-class classifier $\mathbf{z}^{\text{R}} \in \mathbb{R}^4$ for minority-class MI types: Anterior wall MI (AMI), Lateral wall MI (ALMI), Extensive anterior wall MI (ILMI), and Other regions (Other).

The $\mathbf{z}^{\text{HC}}$ first separates normal and abnormal samples. Then the $\mathbf{z}^{\text{M}}$ and $\mathbf{z}^{\text{R}}$ classifiers operate in parallel. During inference, final logits $\hat{\mathbf{y}} \in \mathbb{R}^7$ are computed via soft gating:

$$\begin{aligned} \hat{y}_{\text{HC}} &= z_0^{\text{HC}}, \\ \hat{y}_{\text{IMI}} &= z_0^{\text{M}} \cdot z_1^{\text{HC}}, \quad \hat{y}_{\text{ASMI}} = z_1^{\text{M}} \cdot z_1^{\text{HC}}, \\ \hat{y}_{\text{AMI}} &= z_0^{\text{R}} \cdot z_1^{\text{HC}}, \quad \hat{y}_{\text{ALMI}} = z_1^{\text{R}} \cdot z_1^{\text{HC}}, \\ \hat{y}_{\text{ILMI}} &= z_2^{\text{R}} \cdot z_1^{\text{HC}}, \quad \hat{y}_{\text{Other}} = z_3^{\text{R}} \cdot z_1^{\text{HC}}. \end{aligned} \quad (11)$$

Here, z_1^{HC} represents the predicted probability of the sample being abnormal, while z_i^{M} ($i = 0, 1$) and z_j^{R} ($j = 0, 1, 2, 3$) denote the class-specific activation scores for the i -th majority and j -th minority MI subtypes, respectively. Specifically, z_0^{R} , z_1^{R} , z_2^{R} , and z_3^{R} correspond to AMI, ALMI, ILMI, and Other categories in order. All z values represent normalized class probabilities obtained after the softmax operation in their respective classifier heads.

The final output is obtained by interpolating with a global seven-class classifier $\mathbf{y}^{\text{global}}$ as:

$$\hat{\mathbf{y}}_{\text{final}} = \alpha \cdot \hat{\mathbf{y}} + (1 - \alpha) \cdot \mathbf{y}^{\text{global}}, \quad \alpha = 0.7. \quad (12)$$

Model training uses seven independent binary cross-entropy losses with class-specific weights. This hierarchical structure improves discrimination on minority-class MI types while maintaining robustness on majority-class categories.

IV. EXPERIMENTS

A. Datasets

We evaluate the proposed SPG-Net model on two publicly available benchmark ECG datasets: the PTB-XL and the PTB Diagnostic ECG Database. Both datasets contain standard 12-lead ECG signals along with diagnostic annotations and are widely used in tasks related to MI classification and localization.

The PTB-XL dataset, provided by PhysioNet, consists of 21,837 10-second 12-lead ECG records sampled at 500 Hz. It includes diagnostic labels at multiple levels of granularity. For this study, we extract seven anatomically localized MI-related labels: anterior wall (AMI), inferior wall (IMI), lateral wall (ALMI), posterior wall (PMI), extensive anterior wall (ASMI),

other regions, and normal (HC), and formulate the task as a multi-label classification problem [20].

TABLE I: Class distribution in the subset of the PTB-XL dataset.

Class	Subjects	Records
AMI	348	353
ASMI	2033	2357
ALMI	243	288
IMI	2355	2676
ILMI	416	478
Other (PMI+LMI+IPMI+IPLMI)	278	300
HC (Healthy Controls)	8896	9514
Total	13285	14801

The PTB Diagnostic ECG Database is an earlier standard ECG dataset containing 549 records from 290 subjects, many of which include clearly annotated MI localization labels. Due to its smaller sample size, this dataset is commonly used for small-sample validation and generalization testing [21].

B. Baseline Models

To comprehensively evaluate the effectiveness of SPG-Net, we compare it with five representative deep learning models.

ECG-Transformer [22]: A multi-scale attention-based Transformer that integrates temporal sampling, Squeeze-and-Excitation (SE) blocks, and Transformer encoders to jointly model local and global ECG features. It outperforms CNN-based methods on the PTB-XL dataset in both accuracy and F1 score.

MCA-Net [23]: A multi-task channel-attention model built upon residual backbones with lead-wise attention. By enabling cross-task feature sharing, it achieves over 90% MI detection accuracy on both the PTB and PTB-XL datasets.

ML-ResNet [24]: A ResNet-based ECG classification model with lead-specific branches. It demonstrates high accuracy under both intra-patient and inter-patient evaluation settings for MI localization.

MRM-Net [25]: A multi-label classification model that combines dual-resolution attention with complementary learning to capture both waveform details and rhythm patterns. It achieves state-of-the-art Macro-F1 and AUC scores on the PTB-XL and CPSC2018 datasets.

CBAM [26]: A lightweight 1D attention-based model that incorporates a Convolutional Block Attention Module (CBAM) and compact convolutional encoders. It effectively addresses class imbalance and achieves nearly 99% accuracy on the MIT-BIH dataset, making it suitable for real-time clinical applications.

C. Implementation Details and Evaluation Metrics

All experiments are performed on a server equipped with an Intel Xeon Silver 4114 CPU, 128 GB RAM, and an NVIDIA GeForce GTX 2080 Ti GPU, using PyTorch 1.8.1 with CUDA 10.1 and Python 3.7.12. We adopt AdamW and SGD as optimizers on the PTB-XL and PTB datasets, respectively. The learning rate is initialized to $1e-4$ and the batch size remains

64 throughout. To avoid overfitting, early-stop and dropout strategies are applied in all models and the dropout rate is set to 0.5. The training data is randomly shuffled and the test data is loaded in a fixed order. Pretraining uses only the first fold per dataset to simulate low-resource conditions. For PTB-XL, we perform 10-fold cross-validation; for the smaller PTB dataset, 5-fold cross-validation is used. All results are reported as mean $\pm$ standard deviation to ensure robustness. For the PTB-XL dataset, 90% of the records are used for training, while the remaining 10% are reserved for testing. For the smaller PTB dataset, 80% of the samples are used for training and 20% for testing. We evaluate model performance using five commonly adopted metrics: Subset Accuracy (Accuracy), Macro-F1, AUC, Macro-Precision (Precision), and Macro-Recall (Recall). We adopt macro-averaged metrics to ensure balanced evaluation across classes, especially important under severe class imbalance.

V. RESULTS

A. Comparison on the PTB Dataset

TABLE II: Performance Comparison on the PTB Dataset.

Method	Accuracy (%)	AUC (%)	Macro-F1 (%)	Precision (%)	Recall (%)
ECG-Transformer	34.47 $\pm$ 4.32	37.25 $\pm$ 2.71	35.93 $\pm$ 1.84	23.24 $\pm$ 1.28	85.31 $\pm$ 8.51
MCA-Net	44.38 $\pm$ 4.29	39.65 $\pm$ 3.33	39.02 $\pm$ 3.76	26.66 $\pm$ 3.84	84.03 $\pm$ 7.17
ML-ResNet	44.25 $\pm$ 4.54	39.87 $\pm$ 1.93	38.78 $\pm$ 3.38	26.54 $\pm$ 4.34	82.07 $\pm$ 8.78
MRM-net	34.86 $\pm$ 3.71	33.41 $\pm$ 2.90	35.43 $\pm$ 2.95	23.98 $\pm$ 2.21	73.31 $\pm$ 5.39
CBAM	19.40 $\pm$ 1.75	15.20 $\pm$ 1.02	14.58 $\pm$ 1.61	8.69 $\pm$ 0.92	45.70 $\pm$ 6.36
SPG-Net	45.79 $\pm$ 2.29	44.18 $\pm$ 3.18	41.90 $\pm$ 2.25	29.21 $\pm$ 1.90	80.04 $\pm$ 7.63

As shown in Table II, SPG-Net achieves the highest performance in Subset Accuracy, AUC, Macro-F1, and Macro-Precision, validating the effectiveness of its structural modeling. It surpasses MCA-Net and ML-ResNet in Macro-F1 by 2.9% and 3.1%, respectively.

Although ECG-Transformer attains the highest Macro-Recall (85.31%), it suffers from low precision (23.24%), indicating a strong false positive bias. SPG-Net maintains high recall while significantly improving precision, leading to better F1 balance and more reliable predictions.

Moreover, SPG-Net demonstrates strong robustness under data scarcity and class imbalance, confirming the advantage of its lead graph modeling and multi-path representation framework.

B. Comparison on the PTB-XL Dataset

TABLE III: Performance Comparison on the PTB-XL Dataset.

Method	Accuracy (%)	AUC (%)	Macro-F1 (%)	Precision (%)	Recall (%)
ECG-Transformer	71.18 $\pm$ 2.89	46.82 $\pm$ 3.43	38.43 $\pm$ 4.47	42.26 $\pm$ 8.10	40.32 $\pm$ 3.76
MCA-Net	79.84 $\pm$ 0.74	65.52 $\pm$ 1.53	54.97 $\pm$ 2.21	73.12 $\pm$ 7.90	50.49 $\pm$ 1.96
ML-ResNet	78.71 $\pm$ 0.91	61.41 $\pm$ 2.11	46.84 $\pm$ 2.80	62.79 $\pm$ 7.94	42.35 $\pm$ 2.66
MRM-net	78.52 $\pm$ 1.05	60.34 $\pm$ 2.75	57.49 $\pm$ 1.85	63.91 $\pm$ 2.74	54.24 $\pm$ 1.94
CBAM	76.65 $\pm$ 6.02	54.63 $\pm$ 5.01	38.59 $\pm$ 15.72	44.38 $\pm$ 20.75	36.57 $\pm$ 12.81
SPG-Net	79.92 $\pm$ 0.85	66.18 $\pm$ 2.45	60.62 $\pm$ 2.13	68.59 $\pm$ 2.65	57.03 $\pm$ 3.04

As shown in Table III, SPG-Net achieves the best overall performance in terms of AUC, Macro-F1 Score, Macro-Precision, and Macro-Recall, outperforming all other baseline methods. Notably, SPG-Net outperforms MCA-Net and

MRM-Net in Macro-F1 Score by 5.6% and 3.1%, respectively, and improves Macro-Recall by 6.5% and 2.8%, respectively.

To provide a detailed evaluation of minority-class performance, Table IV presents the recall scores for four under-represented MI categories. As shown in Table IV, SPG-Net demonstrates substantial gains in minority-class recall, particularly for ALMI and ILMI, achieving state-of-the-art sensitivity in these challenging classes. Despite its slightly lower recall on AMI compared to MRM-Net, SPG-Net still maintains competitive performance and provides superior overall class-level stability.

TABLE IV: Minority-Class Recall on PTB-XL Dataset.

Method	AMI Recall (%)	ALMI Recall (%)	ILMI Recall (%)	Other Recall (%)
ECG-Transformer	0.00 $\pm$ 0.00	15.47 $\pm$ 16.57	16.72 $\pm$ 13.93	1.70 $\pm$ 3.30
MCA-Net	5.21 $\pm$ 3.66	44.30 $\pm$ 1.54	4.70 $\pm$ 0.49	4.77 $\pm$ 4.85
ML-ResNet	0.29 $\pm$ 0.90	27.09 $\pm$ 10.99	25.35 $\pm$ 7.82	0.65 $\pm$ 1.35
MRM-net	26.03 $\pm$ 8.17	51.50 $\pm$ 4.67	42.59 $\pm$ 11.47	15.70 $\pm$ 6.57
CBAM	2.86 $\pm$ 4.86	13.45 $\pm$ 20.35	25.03 $\pm$ 24.19	6.18 $\pm$ 9.98
SPG-Net	19.14 $\pm$ 5.63	52.34 $\pm$ 12.36	49.34 $\pm$ 9.68	24.56 $\pm$ 8.95

These results demonstrate that SPG-Net is effective in handling large-scale, imbalanced ECG datasets and robust in detecting minority-class myocardial infarction cases. Its strong performance is attributed to the innovative lead graph construction and multi-path feature learning, which together yield more balanced and accurate classification across a wide range of MI subtypes.

C. Ablation Study on the PTB-XL Dataset

To comprehensively evaluate the contribution of each module in SPG-Net, we conducted an ablation study on the PTB-XL dataset. Specifically, the following variants were compared: (1) SPG-Net w/o CSAC that removes the CSAC module. (2) SPG-Net w/o GSCA that removes the GSCA module. (3) SPG-Net w/o SPM that removes the SPM submodule from GSCA. (4) SPG-Net w/o SAM that removes the SAM submodule from GSCA. (5) SPG-Net w/o TDM that removes the TDM submodule from GSCA. These comparisons aim to assess the impact of each module on overall classification performance and the model’s sensitivity to minority MI categories.

TABLE V: Ablation Results on PTB-XL Dataset.

Model Variant	Accuracy (%)	AUC (%)	Macro-F1 (%)	Precision (%)	Recall (%)
SPG-Net w/o CSAC	79.23 $\pm$ 0.87	64.60 $\pm$ 1.77	60.23 $\pm$ 2.33	67.10 $\pm$ 3.26	57.00 $\pm$ 3.37
SPG-Net w/o GSCA	79.52 $\pm$ 0.92	64.43 $\pm$ 1.43	59.80 $\pm$ 1.94	68.89 $\pm$ 2.63	55.62 $\pm$ 2.85
SPG-Net w/o SPM	79.34 $\pm$ 1.04	64.61 $\pm$ 1.82	60.38 $\pm$ 1.65	68.48 $\pm$ 4.08	56.78 $\pm$ 3.56
SPG-Net w/o SAM	79.72 $\pm$ 0.72	64.02 $\pm$ 1.53	60.87 $\pm$ 2.41	69.35 $\pm$ 2.87	57.01 $\pm$ 3.63
SPG-Net w/o TDM	79.88 $\pm$ 0.84	64.88 $\pm$ 1.53	60.26 $\pm$ 2.25	68.72 $\pm$ 3.79	56.31 $\pm$ 2.96
SPG-Net w/o DBW	79.19 $\pm$ 1.18	64.72 $\pm$ 1.37	60.28 $\pm$ 2.44	69.63 $\pm$ 4.14	56.30 $\pm$ 3.87
SPG-Net (full model)	79.92 $\pm$ 0.85	66.18 $\pm$ 2.45	60.62 $\pm$ 2.13	68.59 $\pm$ 2.65	57.03 $\pm$ 3.04

As shown in Table V, removing either the CSAC or GSCA module consistently reduces Macro-F1 and Macro-Recall, confirming their importance for class-specific discrimination. GSCA captures static anatomical priors and temporal dependencies, and its removal weakens spatial-temporal modeling for rare MI types. Disabling any single GSCA submodule also leads to performance drops, especially in Macro-Recall. The SPM encodes prior-guided structure, the SAM adapts to patient-specific lead patterns, and the TDM models waveform

dynamics. These results demonstrate that all SPG-Net components contribute synergistically to robust classification under label imbalance.

VI. DISCUSSION

A. Effectiveness of the DBW Module

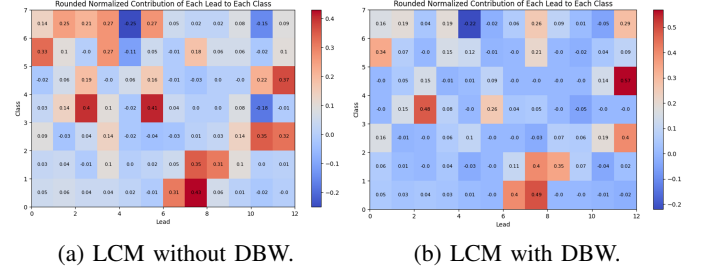


Fig. 4: Visual comparison of LCMs with and without DBW.

Fig. 4 compares the lead contribution maps (LCMs) before and after applying the DBW module. After incorporating DBW, lead contributions become more focused and consistent across samples. In particular, clinically critical leads such as Lead III and Lead V6 exhibit enhanced and stable activation across multiple MI classes, aligning well with domain knowledge.

Moreover, the difference between specific and non-specific lead contributions becomes more pronounced. In the DBW-enhanced LCM (Fig. 4, right), the maximal lead-class contribution values (e.g., Lead V6 for ILMI: 0.57) are significantly higher than those in the non-DBW variant (Fig. 4, left: 0.37), while non-relevant leads are further suppressed (e.g., contributions close to zero or negative). This increased disparity improves the signal-to-noise ratio in gradient backpropagation and allows the model to focus on more discriminative features.

These changes suggest that DBW effectively reweights class gradients, making lead-specific saliency maps sharper and more class-discriminative. As a result, minority-class MI types benefit from improved attention to subtle yet clinically relevant ECG cues, enhancing the overall sensitivity and robustness of the classifier.

B. Impact of Lead Graph Construction Methods

To further evaluate the impact of lead graph construction strategies on model performance, comparative experiments were conducted using three types of static graphs embedded within the SPG-Net framework:

- Prior Graph: A hand-crafted lead connection graph based on clinical anatomical knowledge, used as a fixed topology [27].
- LCM Sparse Graph (PTB): A class-aware sparse structural graph generated from LCMs using the PTB dataset.
- LCM Sparse Graph (PTB-XL): A class-aware sparse graph derived from LCMs on the PTB-XL dataset and transferred for use in the PTB dataset.

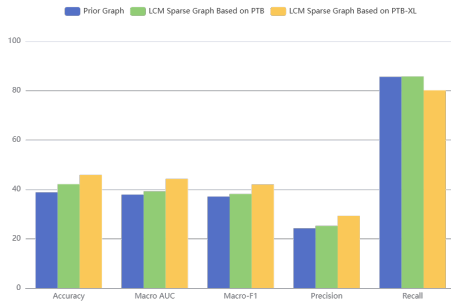


Fig. 5: Experimental Results with Different Lead Graph Construction Methods.

As shown in Fig. 5, LCM-based sparse graphs consistently outperform manually designed priors. Using PTB-XL-derived LCM graphs improves Macro-F1 from 38.07% to 41.90% and Macro-Precision from 25.15% to 29.21%. These results indicate that data-driven structural graphs built from large-scale datasets offer better generalization. This supports the use of cross-dataset spatial priors for robust MI localization under diverse clinical conditions.

VII. CONCLUSION

In this study, we proposed SPG-Net, a lead contribution-guided framework that effectively improves ECG-based MI localization under class-imbalanced conditions. The model demonstrated superior performance and interpretability on benchmark datasets. Future work will focus on enhancing recognition of extremely rare classes, improving efficiency, and advancing clinical applicability.

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